



Methane rhizospheric oxidation and its ecosystem significance across different aquatic plant species

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Abstract Vegetated inland water habitats are important global emitters of methane (CH_4), but the variability in their CH_4 emission is poorly understood, making estimates and predictions uncertain. Methane is believed to be primarily released into the atmosphere via plant lacunal tissues in dense vegetated zones, although experiments have also shown that a large fraction of CH_4 can be consumed in the rhizosphere before it diffuses into the roots. Methane rhizospheric oxidation measurements have been limited to few species in situ, and it is still unclear how it differs between species. To determine the implications of CH_4 rhizospheric oxidation for plant- CH_4 release, I measured CH_4 rhizospheric oxidation and transport rates for 11 plant species with different gas transport efficiencies using dual isotopes in situ. Overall, the fraction of CH_4 produced that was oxidized was higher for plants that transport gases efficiently (up to 81%), and was negatively correlated to plant CH_4 release. These findings suggest that CH_4 oxidation decreases CH_4 release from plants that

efficiently transport gases, although further studies are needed to assess the causality of this relationship. Such mechanistic understanding of plant- CH_4 release is necessary to define aquatic plant functional types with contrasting CH_4 emissions, and to reduce uncertainties in estimates of CH_4 freshwater emissions.

Keywords Macrophytes · Greenhouse gas flux · Plant-mediated transport · Aerenchyma · Methanotrophy · Convective-bulk flow · Dual stable isotopes

Introduction

Wetlands and inland waters release large amounts of methane (CH_4), a gas with a global warming potential 27–30 times that of carbon dioxide (CO_2) at a 100 yr-scale (Canadell et al. 2021). Accordingly, these ecosystems could account for over one third of global CH_4 emissions (Watts et al. 2026). In particular, vegetated aquatic freshwater habitats, such as littoral zones of lakes, ponds and inland wetlands, are hotspots of CH_4 emission, but CH_4 emissions in these habitats are highly variable and, to date, poorly understood (Rosentreter et al. 2021; Bodmer et al. 2024).

The large contribution of vegetated aquatic habitats to CH_4 emission arises partially due to the extremely high productivity of aquatic plants, which sometimes reaches that of vegetation in the rainforest (Wetzel 2001) and generates high quantities of macrophyte

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detritus that can fuel methanogenesis (Whiting and Chanton 1993; Grasset et al. 2021). However, the proportion of CH_4 produced that ultimately reaches the atmosphere is still highly uncertain and depends on how much CH_4 is oxidized before or during CH_4 transport via different pathways, e.g., the diffusion of CH_4 through the water/air interface, the release of CH_4 as bubbles, known as ebullition, or the transport of CH_4 within plant tissues (Bastviken 2009).

Emergent and floating plants directly transport CH_4 from the sediment to the atmosphere within their lacunal tissue called “aerenchyma”, whose primary function is to supply oxygen (O_2) to the roots (Dacey 1980; Brix et al. 1992; Colmer 2003). Some plant species, hereafter called “nonefficient plants”, passively transport gases within the aerenchyma via diffusion, whereas others, hereafter called “efficient plants”, can generate a convective bulk flow to transport gases during the day (Dacey 1980; Colmer 2003; Vroom et al. 2022). This convective bulk flow mainly originates from humidity differences inside and outside the leaves that dilute nitrogen gas (N_2) and O_2 concentrations inside the leaves. Because of this concentration gradient, N_2 and O_2 diffuse into young “influx” leaves or stems and generate a pressure gradient (Colmer 2003; van den Berg et al. 2020). The internal gas mixture then flows through the rhizome, where it is enriched in CH_4 and generally exits older “efflux” leaves or stems (Chanton et al. 1993).

Compared with the other pathways of CH_4 release, the release of CH_4 by plants is considered a predominant pathway and could be responsible for up to 90% of emissions in dense vegetated habitats during the productive season (Holzapfel-Pschorn et al. 1986; Van Der Nat and Middelburg 1998a; van den Berg et al. 2020). However, high rates of CH_4 oxidation can occur in the rhizosphere (i.e., rooting zone) of aquatic plants because O_2 leaks from the roots (i.e., radial oxygen loss) and can be used by methanotrophs to oxidize CH_4 (Colmer 2003). Rhizospheric oxygenation can also suppress CH_4 production in soils and sediments with low reactivity by replenishing alternative electron acceptors (Agethen et al. 2018). Experiments (Calhoun and King 1997; Ström et al. 2005; Laanbroek 2010) and in situ measurements, conducted for limited plant species and mostly in fens and paddy fields (Tyler et al. 1997; Popp et al. 1999; Zhang et al. 2016), have indicated that 20–100% of the produced CH_4 can be oxidized in the plant

rhizosphere. Such numbers vary widely among plant species and are potentially related to their root systems and efficiencies in transporting O_2 to the rhizosphere (Armstrong et al. 1996; Ström et al. 2005). The latter theory is partially supported by the fact that plants that transport gases efficiently generally do not show higher CH_4 emissions than other plants (Kao-Kniffin et al. 2010; Vroom et al. 2022; Bodmer et al. 2024). As the studies that have measured CH_4 rhizospheric oxidation were mostly experimental and featuring only few species, it is still unclear whether in situ CH_4 rhizospheric oxidation differs between nonefficient and efficient aquatic plants, and how it relates to CH_4 release rates.

During CH_4 oxidation, lighter isotopes are preferentially used, leaving the remaining CH_4 enriched in heavier isotopes (^{13}C and ^2H). The $\delta^{13}\text{C}$ of CH_4 is often used to calculate the fraction of newly produced CH_4 that has been oxidized in the water column or the in rhizosphere (Bastviken et al. 2002; Zhang et al. 2016). However, plant CH_4 transport by diffusion through the aerenchyma results in a similar ^{13}C enrichment, making it difficult to disentangle the effects of oxidation and plant transport on the $\delta^{13}\text{C}$ of CH_4 contained in plant aerenchyma. The use of the $\delta^2\text{H}$ of CH_4 allows to better constrain the extent of CH_4 oxidation because the fractionation of $\text{H}-\text{CH}_4$ due to oxidation or diffusion within the aerenchyma differs by a factor greater than 10, ca. 100–200‰ vs. ca. 10‰ for oxidation and diffusion, respectively (Chanton 2005; Tsunogai et al. 2020). Until now, dual isotopes of CH_4 (i.e. the combined use of $\delta^2\text{H}$ and $\delta^{13}\text{C}$) have been used to calculate CH_4 oxidation in the water column (Einzmann et al. 2022) but not in the rooting zones of aquatic plants.

In this study, I quantified CH_4 oxidation with dual isotopes in the rhizospheres of 11 aquatic plant species with different transport efficiencies and related CH_4 rhizospheric oxidation to both plant CH_4 transport and sediment CH_4 concentration. I hypothesized that a greater fraction of the CH_4 produced is oxidized in the rhizosphere of plants that efficiently transport CH_4 than in nonefficient plants. Furthermore, I expected that greater rhizospheric oxidation of CH_4 would generally decrease plant CH_4 transport.

Materials and methods

Sampling design

During the summer of 2023, CH₄ emissions, CH₄ concentration and isotopic composition from plant aerenchyma and sediment pore water samples were measured for 11 different plant species (Table 1) in two lakes: Ånnsjön (63.2575 N, 12.5589 E, end of June) and Strandsjön (59.8681 N, 17.1711E, end of August). Additionally, during the early autumn of 2020 (beginning of September) in Strandsjön, CH₄ concentration and isotopic composition from plant aerenchyma and sediment pore water samples were measured for 3 of the 11 species sampled in 2023 (Table 1). Plants growing in both shallow (0.2–0.5 m)

and relatively deep littoral zones (0.5–1.5 m) were selected because the efficiency of plant gas transport differs between species colonizing different lake habitats (Tornberg et al. 1994; Strand 2002; Vroom et al. 2022). The selected species belonged to different growth forms, i.e., emergent and floating leaved plants anchored to the sediment, and functional types, i.e., forbs and graminoids including sedges and grasses (Table 1). For each plant species, four to five mature individuals or groups of individuals (*Caltha palustris*, *Eleocharis palustris*, *Phragmites australis*) were selected based on their size and the leaf color, from either the same vegetation patch or nearby patches. Methane emissions were measured above all individuals or groups of individuals depending on their size, along with diffusive CH₄ emissions and

Table 1 List of the sampled species

Genus/species	Category (growth form and/or type)	Abbreviation	Site and year of collection	Transport efficiency*	References
<i>Caltha palustris</i>	Emergent-Forb	Calt	Ånnsjön, 2023	Non-efficient**	Ding et al. (2005), Brix et al. (1992) for Cyperaceae
<i>Carex</i> sp.	Emergent-Sedge	Carex	Ånnsjön, 2023	Non-efficient**	
<i>Iris pseudacorus</i>	Emergent-Forb	Iris	Strandsjön, 2023	Non-efficient	Strand (2002)
<i>Sagittaria sagittifolia</i>	Emergent-Forb	Sagi	Strandsjön, 2023	Non-efficient**	Sebachner et al. (1985) for <i>Sagittaria</i> genus
<i>Schoenoplectus lacustris</i>	Emergent-Sedge	Scir	Strandsjön, 2023	Efficient [†]	Strand (2002)
<i>Eleocharis palustris</i>	Emergent-Sedge	Eleo	Ånnsjön, 2023	Efficient [†]	Strand (2002)
<i>Phragmites australis</i>	Emergent-Grass	Phra	Strandsjön, 2020 and 2023	Efficient	Strand (2002)
<i>Typha latifolia</i>	Emergent-Forb	Typh	Strandsjön, 2020 and 2023	Efficient	Strand (2002)
<i>Nuphar lutea</i>	Floating anchored to the sediment	Nuph	Strandsjön, 2020 and 2023	Efficient	Dacey (1980); Grosse et al. (1991)
<i>Nymphaea alba</i>	Floating anchored to the sediment	Nymp	Strandsjön, 2023	Efficient	Grosse et al. (1991)
<i>Potamogeton natans</i>	Floating anchored to the sediment	Pota	Strandsjön, 2023	Efficient	Heilman and Carlton (2001) for <i>Potamogeton</i> genus, Grosse et al. (1991) for floating leaved species

*Transport efficiency was assessed from gas flow rate measurements unless specified and marked by **. Nonefficient plants have gas flow rates non detectable or < 0.1 cm³ min⁻¹ culm or petiole⁻¹ while efficient plants have rates > 0.1 cm³ min⁻¹ culm or petiole⁻¹ (Strand 2002)

***Caltha palustris* is assumed to be a nonefficient gas transporter because it is colonizing shallow waters (Vroom et al. 2022). *Carex* species are commonly assumed to be nonefficient gas transporters (see references). No detectable flow rates were measured for several species of the *Sagittaria* genus (Sebachner et al. 1985)

[†]Low flow rates are measured despite these plants possessing an efficient mechanism of gas transport (Strand 2002)

CH₄ concentration and stable isotope composition in plant aerenchyma and in the pore water of nearby sediment.

Methane flux measurements

Methane emissions were measured using a floating chamber placed above water both with and without plant parts for 5–7 min to limit increases in humidity and temperature inside the chamber (Knapp and Yavitt 1992). Diffusive CH₄ emissions were measured in water adjacent to each species, which was possible because low-density plant stands were preferentially sampled. Measurements were taken between 10:00 and 15:00, with most being taken between 11:00 and 13:00; the full sampling period allowed one species to be sampled per day. To ensure pressure equilibration when placing the chambers above water, the valve was only closed after the chamber was put in place. Increases in the gas concentration in the chamber were measured by collecting 10–20 mL of gas every minute with a syringe, leading to 5 samples for each chamber measurement. The gas samples were stored for a maximum of 24 h before they were analyzed using a custom-made circuit connected to a greenhouse gas analyzer (Ultraportable Greenhouse Gas Analyzer, Los Gatos Research; Grasset et al. 2018). Both plant and water fluxes were reported per surface area of water in mg CH₄ m⁻² h⁻¹. Plant CH₄ transport fluxes were positive if CH₄ was released or negative when CH₄ was taken up by plants. Different sizes of transparent chambers were chosen according to the size of the species (Fig S1). The two tallest flexible chambers were made from gas-tight Tedlar Sampling Bag (Keika Ventures, Chapel Hill, NC 27515). One was used for *Phragmites australis*, *Typha Latifolia* and *Schoenoplectus lacustris*, and had a height of 2 m and a full volume of 161 L when not squeezed. The other flexible chamber was used for *Caltha palustris*, *Carex* sp. and *Iris pseudacorus*, and had a height of 1 m and a full volume of 85 L. These chambers were gently squeezed just below the water surface around the plant stems to measure the flux only related to plant transport (called “plant-only flux” hereafter), and were equipped with a fan to ensure fast air mixing inside the chamber. The height at which the chamber was squeezed was noted to calculate the effective volume of the chamber. The plant flux was calculated relative to the full surface

area of the chamber, assuming the plants present in the squeezed chambers were the only plants that could be contained within the non-squeezed chamber. This approximation was valid since low-density plant stands were sampled. The two smallest rigid chambers were made with inverted polypropylene buckets and were used for adjacent water measurements (called “water-only flux”) as well as measurements above small emergent and floating-leaved plants. The chamber used for *Eleocharis palustris*, *Sagittaria sagittifolia* had a height of 21 cm and volume of 10 L. The chamber used for *Nuphar lutea*, *Nymphaea alba* and *Potamogeton natans* had a height of 13 cm and a volume of 14 L. For these species, the flux related to plant transport was calculated by subtracting water adjacent measurements from plant and water measurements. The leaf blades floating at the water surface and enclosed during the plant and water measurements were brought back to the laboratory, and their surface areas were calculated to infer the exact surface area of the water in the chamber. For *Nuphar lutea* and *Nymphaea alba*, only one leaf blade was enclosed in the chamber. I may have sampled relatively more influx or efflux leaves, which could lead to under- or overestimation of plant CH₄ release (Text S1). The two buckets were not fully transparent and decreased the photosynthetically active radiation (PAR) by < 12% under sunny conditions; however, such a small decrease should not have affected plant gas transport (Armstrong and Armstrong 1990).

A linear regression between CH₄ concentration and time was modeled for each measurement to calculate the plant and water-only fluxes. The regressions were visually observed, and measurements were discarded when the concentration sharply increased, likely due to ebullition, and when initial concentrations in the chambers were overly high (> 8 to 10 ppm). These high initial concentrations likely occurred because of sediment disturbance or release of gas bubbles trapped below plant leaves when placing the chamber. For water-only fluxes, 8 out of the 49 measurements were removed, and for plant-only fluxes, 7 out of the 49 measurements were removed (Table S1). Some measurements that were kept had a low R² (< 0.6 for 6 water-only fluxes and 11 plant only fluxes) because of low CH₄ emissions or even uptake (n = 6 plant only-fluxes), as recommended by Jentsch et al. (2025).

Gas sampling in plant aerenchyma

Methane lacunal concentrations were measured for three of the five individuals or groups of individuals above which CH₄ emissions were measured. Plant lacunal gas samples were collected directly after the plant flux measurements by inserting a needle connected to a syringe inside the plant stem/petiole just below the water surface and gently pulling the syringe. When it was not possible to obtain approximately 2–3 mL of gas sample from one individual, gas samples from several individuals were pooled. The samples were transferred into 12 mL gas-tight vials (Soda Glass Vials, Labco, High Wycombe, UK) that were previously flushed and filled with N₂ (Sturm et al. 2015). A small subsample (0.5 mL) was analyzed with the greenhouse gas analyzer to determine whether further dilution was needed before stable isotope analysis. The samples were divided into two vials, one for δ²H and one for δ¹³C, and analyzed for CH₄ stable isotopic composition and concentration using a Thermo Scientific GasBench-PreCon coupled to a Delta V Plus isotope ratio mass spectrometer at the Stable Isotope Facility of the University of California (Davis, California, U.S.A.). Stable isotope results are reported in δ units (‰) relative to the VPDB (Vienna Pee Dee belemnite) and VSMOW (Vienna Standard Mean Ocean Water) standards for δ¹³C and δ²H, respectively, as follows:

$$\delta = (R_{\text{sample}}/R_{\text{standard}} - 1) \times 1000 \quad (1)$$

where R is the ¹³C/¹²C or ²H/¹H ratio measured in the sample and/or the standard.

For efficient plants, CH₄ isotopic signals and concentrations strongly change between night and day and during the transition phases because plants switch their transport mechanisms from diffusion at night to convective bulk flow during the day (Chanton et al. 2002). As samples were collected between 10:00 and 15:00, with most collected at 12:00, they were sampled several hours after or before these transition phases. In Uppland (Strandsjön) at the end of August, the sun rises at 5:30, and in Jämtland (Ånnsjön) at the end of June, the sun rises at 3:00.

Sampling of CH₄ dissolved in sediment pore water

For all species sampled in 2023 and *Nuphar lutea* sampled in 2020, dissolved CH₄ in sediment pore

water was sampled from three sediment cores collected outside the bulk rhizosphere, either between plants present at a low density or next to the plant stands. The location for sediment sampling was relatively close to plants to reflect the isotopic signature of CH₄ produced from plant detritus. The sediments contained few or no roots and were assumed to be predominantly anoxic. Sediment cores were sampled by manually pushing into the sediment a 1 m-long acrylic tube with pre-drilled holes covered with tape following Donis et al. (2017). In 38 mL bottles, 5 mL of lake water was added and the bottles were left open until the addition of sediment. 2 mL of sediment was sampled from each core by inserting a 3 mL headless syringe into the pre-drilled holes at a 10 ± 1 cm depth, and immediately added to the bottles. The bottles were then closed with gas tight septa and the sediment was equilibrated with lake water and atmospheric air by shaking the bottles for one minute. A total of 15 mL of headspace was then transferred into a pre-evacuated 12 mL vial. One control bottle per species, not spiked with 2 mL of sediment, was also sampled to quantify CH₄ contained in the lake water and atmospheric air.

In addition, for Strandsjön, sediment gas samplers were used to sample CH₄ dissolved in the sediment pore water next to the plant roots, i.e., in the rhizosphere. The sediment gas samplers were built similarly to those in the literature using permeable silicone tubing between 10 and 15 cm below the sediment surface connected to a gas-tight polyurethane tube and a three-way valve (Estop-Aragónés et al. 2012; Praetzel et al. 2020). Six sediment gas samplers were inserted at the end of June 2023: three samplers in patches of *Nuphar lutea* and at the edge of *Phragmites australis*, and three samplers in *Phragmites australis* stands. Approximately 3 mL of gas was collected at the end of August from each gas sampler and transferred into 12 mL gas-tight vials that had previously been flushed and filled with N₂.

The samples were divided into two vials for CH₄ stable isotopic composition (one vial for δ²H and one vial for δ¹³C) and concentration at the Stable Isotope Facility, and most of the samples were further diluted before being sent. The CH₄ concentration in ppm in the gas sample was converted into μmol L⁻¹ to obtain the CH₄ concentration in the sediment dissolved pore water using Henry's law (Yamamoto et al. 1976).

Calculation of the fraction of oxidized CH₄

The fraction of newly produced CH₄ that was oxidized in the rhizosphere, F , was calculated by comparing the isotopic signature of CH₄ ($\delta^{13}\text{C}$ and $\delta^2\text{H}$) before and after oxidation. This fraction corresponds to the amount of CH₄ removed by oxidation divided by the initial amount of CH₄ before oxidation. The isotopic signature of CH₄ after oxidation (δ_{final}) was derived from CH₄ sampled in the plant aerenchyma while the isotopic signature of CH₄ before oxidation (δ_{initial}) was estimated from CH₄ sampled in the sediment outside the bulk rhizosphere. Similarly, Tyler et al. (1997); Popp et al. (1999) and Zhang et al. (2016) evaluated CH₄ oxidation by comparing the signal of $\delta^{13}\text{C}$ -CH₄ released by plants with that of soils in peatlands and rice paddies; however, to my knowledge, this is the first study to use both $\delta^2\text{H}$ and $\delta^{13}\text{C}$ for this calculation. F was calculated according to the Rayleigh model for both closed (F_{closed} , Eq. 2; Liptay et al. 1998) and open systems (F_{open} , Eq. 3; Happell et al. 1994) because the rhizosphere does not fully fulfill the assumption of either a closed or an open steady-state system:

$$\ln(1 - F_{\text{closed}}) = (\ln(\delta_{\text{initial}} + 1000) - \ln(\delta_{\text{final}} + 1000)) / (\alpha_{\text{ox}} - 1) \quad (2)$$

$$F_{\text{open}} = (\delta_{\text{final}} - \delta_{\text{initial}}) / ((\alpha_{\text{ox}} - 1) \times 1000) \quad (3)$$

α_{ox} is the isotope fractionation factor due to the microbial oxidation of CH₄. The fractionation factor for C-CH₄, $\alpha_{\text{ox,C}}$, has been extensively measured and often converges at approximately 1.02 (Bastviken et al. 2002; Zhang et al. 2016), which is the value I used for my calculations. The fractionation factor for H-CH₄, $\alpha_{\text{ox,H}}$, has been measured less frequently but could be calculated from the ratio of stable hydrogen vs. stable carbon isotope fractionation, i.e. $\epsilon_{\text{ox,H}}$ vs. $\epsilon_{\text{ox,C}}$ with:

$$\epsilon_{\text{ox}} = (\alpha_{\text{ox}} - 1) \times 1000 \quad (4)$$

This ratio is approximately 12 ± 4 according to Tsunogai et al. (2020), and I chose the value 9.3 given in Einzmann et al. (2022) which resulted in $\alpha_{\text{ox,H}} = 1.186$.

The δ_{initial} value used was the signal of CH₄ dissolved in sediment pore water sampled outside the bulk rhizosphere and close to the respective plant

species for the 2023 dataset. In 2020, sediment was collected only close to the patch of *Nuphar lutea* and used to calculate oxidation for the other plants sampled during the same period (i.e., *Phragmites australis* and *Typha latifolia*).

The δ_{final} value was derived from that measured in the aerenchyma of the plant stem or petiole (δ_{measured}). To do so, δ_{final} was corrected for the discrimination that occurs during the transport of the gas at the entry (sediment/root interface) or exit (stem or petiole/atmosphere interface) of the plant as follows:

$$\delta_{\text{final}} = \delta_{\text{measured}} + \epsilon_{\text{plant transport}} \quad (5)$$

where $\epsilon_{\text{plant transport}}$ is the fractionation that occurs at the entry and/or exit of the plant. For plants that use convective bulk flow, I used $\epsilon_{\text{plant transport}} = 1\text{‰}$ because a slight decrease in $\delta^{13}\text{C}$ of approximately 1‰ is generally observed when sediment vs. aerenchyma signals are compared, potentially because of the slower diffusion of the heavier isotope through the sediment/root interface (Chanton and Whiting 1996; Chanton et al. 2002). The same shift is expected for $\delta^2\text{H}$ because ^{13}C -CH₄ and ^2H -CH₄ have the same mass (Chanton and Whiting 1996; Chanton 2005).

For plants that employ diffusive transport, I used $\epsilon_{\text{plant transport}} = -10.6\text{‰}$. For these plants, increases in $\delta^{13}\text{C}$ -CH₄ and $\delta^2\text{H}$ -CH₄ in the plant aerenchyma of approximately 10.6 + to 3.7‰ relative to rhizospheric CH₄ signal are generally observed (Chanton and Whiting 1996; Chanton 2005) because the heavier isotope is more slowly released to the atmosphere and thus accumulates in the plant aerenchyma.

With this method, the precise location of CH₄ oxidation cannot be determined, as oxidation is measured between the sediment and the exit of the plants, and it is possible that some CH₄ oxidation occurs within plant tissues (Calhoun and King 1997; Jeffrey et al. 2021). However, most oxidation is assumed to occur in the rhizosphere (Tyler et al. 1997; Popp et al. 1999; Zhang et al. 2016; Guo et al. 2024) and we refer to oxidation as “rhizospheric oxidation” in the text.

The mean fraction of oxidized CH₄ was calculated using $\delta^2\text{H}$ and $\delta^{13}\text{C}$ -CH₄, and set to 0 when calculations based on either $\delta^{13}\text{C}$ or $\delta^2\text{H}$ yielded a negative value, which indicates that either the values used for

calculations were not accurate, or other mechanisms took place.

Statistical analyses

All statistical analyses were performed using the R software (R Core Team 2020), with the statistical significance defined as p value < 0.05 . A non-parametric Wilcoxon test was used to test if the fraction of oxidized CH_4 differed between efficient and non-efficient plants. To test if greater rhizospheric oxidation of CH_4 would generally decrease plant CH_4 transport, correlations were tested between the CH_4 concentration in the sediment pore water, the fraction of oxidized CH_4 and the CH_4 plant and water fluxes for the dataset collected in 2023. The nonparametric Spearman test adjusted for ties (cor.test function) was used to test all correlations because of the low number of observations and the presence of ties (i.e. several observations sharing the same value). All replicate measurements were used to test correlations between the fraction of oxidized CH_4 and CH_4 fluxes, since measurements were made for the same individuals or groups of individuals ($n = 27$). The correlations between dissolved CH_4 in sediment pore water, CH_4 oxidation and CH_4 fluxes were tested after averaging values for each species ($n = 11$ observations) because measurements of dissolved CH_4 in sediment pore water were not specifically paired with flux or aerenchyma measurements.

Results

Methane concentration and stable isotopic signature in sediment and plant aerenchyma

For the sediment sampled close to emergent plants by coring, CH_4 dissolved in the sediment pore water had a mean $\delta^{13}\text{C}$ of -60‰ (min of -66 , max of -54) and a mean $\delta^2\text{H}$ of -340‰ (min of -370 , max of -299). The sediment sampled close to the *Nuphar lutea* stand in 2020 had a relatively similar CH_4 isotopic signature, with a mean $\delta^{13}\text{C}$ of -64‰ (min of -66 , max of -61) and a mean $\delta^2\text{H}$ of -319‰ (min of -331 , max of -304). However, for the sediment sampled close to floating plants in 2023, the $\delta^2\text{H}$ of CH_4 was higher (mean of -284‰ , min of -317 ,

max of -264), and the $\delta^{13}\text{C}$ markedly lower (mean of -70‰ , min of -74 , max of -66 ; Figs. 1, S2).

Dissolved CH_4 sampled in the rhizosphere with the sediment gas sampler had relatively high mean values of $\delta^{13}\text{C}$ (-55‰ ; min -60 , max -52) and $\delta^2\text{H}$ (307‰ ; min -331 , max -274), indicating that some of the CH_4 initially produced had been lost by oxidation (Text S2) (Fig. 1).

The $\delta^{13}\text{C}$ and $\delta^2\text{H}$ values of CH_4 in the aerenchyma were almost always higher than values of CH_4 in the sediment pore water samples, indicating discrimination due to either plant transport or biological oxidation. For several efficient plants, the increase in $\delta^2\text{H}$ was generally higher than the increase in $\delta^{13}\text{C}$ (85‰ vs 8‰ in average for *Eleocharis palustris*, *Phragmites australis* and the floating-leaved plants; Fig. S2), indicating that oxidation was causing the shift in isotopic signal. However, *Potamogeton natans* and *Phragmites australis* sometimes presented signals

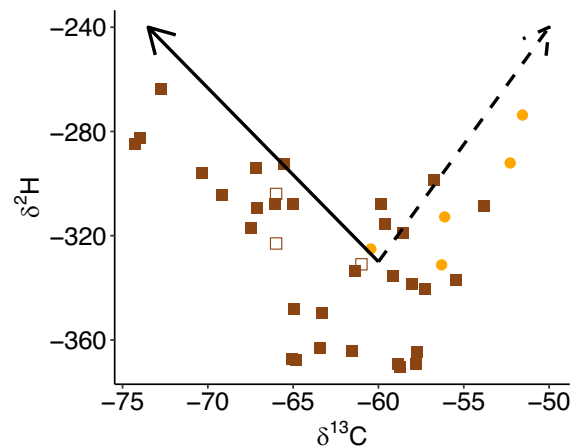


Fig. 1 Stable isotopic signatures of CH_4 ($\delta^{13}\text{C}$ and $\delta^2\text{H}$ in ‰) dissolved in sediment pore water obtained by sampling sediment cores between plants or close to emergent plant stands (brown squares) and by performing equilibration in a sediment gas sampler placed in the rhizosphere (orange dots). Samples from 2023 appear in brown full squares and samples for 2020 (for *Nuphar lutea* only) in empty squares. Two samples from *Eleocharis palustris* stands are excluded because the CH_4 concentrations were too low to be detected. Two arrows show how the signal of CH_4 produced from acetate fermentation, starting with typical values of $\delta^{13}\text{C} = -60\text{‰}$ and $\delta^2\text{H} = -330\text{‰}$, shifts toward more predominant CH_4 originating from CO_2 reduction (full line toward a theoretical signal of $\delta^{13}\text{C} = -75\text{‰}$ and $\delta^2\text{H} = -230\text{‰}$) or when oxidized (dashed line toward a theoretical signal of $\delta^{13}\text{C} = -40\text{‰}$ and $\delta^2\text{H} = -150\text{‰}$), as in Douglas et al. (2021). For plant name abbreviations see Table 1

in aerenchyma substantially lower than in sediment (difference greater than 5 ‰ for $\delta^{13}\text{C}$; Fig. S2).

Both the concentration of CH_4 in dissolved pore water and that in aerenchyma varied widely across species, ranging from several orders of magnitude, from 6 to 2992 $\mu\text{mol L}^{-1}$ for sediment and from 2 to 14,800 ppm for aerenchyma (Fig. S3). Additionally, CH_4 concentrations in the sediment pore water of the two samples collected close to the *Eleocharis palustris* stands were too low to be detected. The concentrations in plants with efficient gas transport were sometimes very low, e.g., less than 300 ppm for *Nuphar lutea*, *Phragmites australis*, *Potamogeton natans*, or *Typha latifolia* or high, e.g., greater than 5000 ppm for *Nymphaea alba*, *Typha latifolia*, or *Nuphar lutea*, whereas the concentrations in non-efficient plants were always above 300 ppm and often above 1000 ppm.

Methane oxidation

When a closed-system model was used, up to 89% of the produced CH_4 was oxidized in the rhizosphere of efficient plants (*Eleocharis palustris*, *Nuphar lutea*, *Nymphaea alba*, *Potamogeton natans*, *Phragmites australis*, and *Typha latifolia*) (Fig. 2). The highest value of oxidized CH_4 was obtained for *Nuphar lutea*,

from calculations with $\delta^{13}\text{C}$ only. The mean fraction of oxidized CH_4 calculated using $\delta^2\text{H}$ and $\delta^{13}\text{C}$ - CH_4 was significantly higher for efficient plants, irrespective if the mean value was set to 0 when either $\delta^{13}\text{C}$ or $\delta^2\text{H}$ yielded a negative value (Fig. 2) or not (Fig. S4). The mean fractions of oxidized CH_4 were 0.05–0.36 for *Eleocharis palustris*, 0–0.81 for *Nuphar lutea*, 0.06–0.34 for *Nymphaea alba*, 0–0.65 for *Potamogeton natans*, 0–0.43 for *Phragmites australis*, and 0–0.25 for *Typha latifolia*. For these plants, the value of the fraction of CH_4 oxidized was generally consistent between calculations based on $\delta^2\text{H}$ and $\delta^{13}\text{C}$, with a difference below 0.23. However, for two samples of *Potamogeton natans* and one sample of *Phragmites australis*, which had very low $\delta^{13}\text{C}$ values in the aerenchyma (−80, −77 and 70‰, respectively), the difference between the fraction of CH_4 oxidized based on $\delta^2\text{H}$ and $\delta^{13}\text{C}$ calculations was greater than 0.23. For the non-efficient plants and *Schoenoplectus lacustris*, which has a low transport efficiency (Table 1), the fraction of CH_4 oxidized did not exceed 0.37, and calculations based on $\delta^2\text{H}$ and $\delta^{13}\text{C}$ were often not consistent, with either $\delta^{13}\text{C}$ or $\delta^2\text{H}$ rendering a negative value.

The use of the open system model resulted in similar patterns but led to less conservative values, sometimes substantially deviating from the theoretical

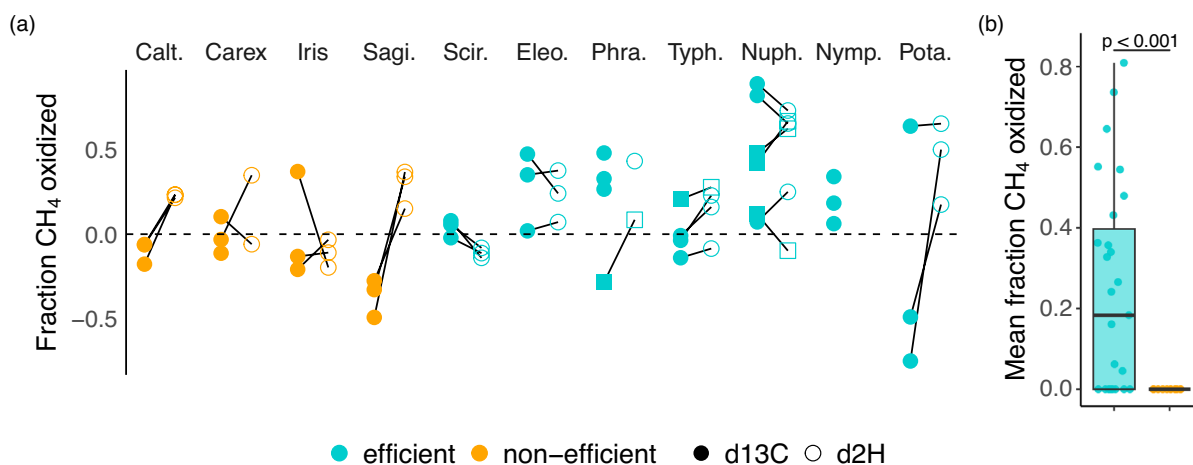


Fig. 2 Fraction of oxidized CH_4 in the rhizosphere for the efficient gas-transporting plants in blue and the non-efficient transporters in orange, **a** based on $\delta^{13}\text{C}$ - CH_4 (dots) and $\delta^2\text{H}$ - CH_4 (circles) and **b** calculated as a mean using $\delta^{13}\text{C}$ and $\delta^2\text{H}$ calculations. In panel **a** the lines connect the $\delta^2\text{H}$ and $\delta^{13}\text{C}$ values of CH_4 originating from the same sample. Data collected in Sep-

tember 2020 for a subset of the species are included in the full ($\delta^{13}\text{C}$) and empty ($\delta^2\text{H}$) squares. For plant name abbreviations see Table 1. In panel **b** the mean was set to 0 when either $\delta^{13}\text{C}$ or $\delta^2\text{H}$ yielded a negative value. The mean fraction of oxidized CH_4 significantly differed between efficient and non-efficient plant as denoted by the p-value (Wilcoxon test)

range of 0–1; the fraction of oxidized CH_4 for *Nuphar lutea* was approximately 2 (Fig. S5). Only the values obtained from the closed-system model were thus kept for the correlation analyses. In addition, I calculated CH_4 oxidation using the same average value for sediment pore water to assess how the results changed when different sediment signals were used for calculations (Text S2). In this case, the fraction of oxidized CH_4 calculated was relatively similar for most species but decreased to ca. 0 for *Nymphaea alba* and increased to ca. 0.1–0.3 for *Schoenoplectus lacustris* (Fig. S6).

Methane fluxes above plants and water surface

Fluxes measured above plants greatly varied among species, while water-only flux (diffusive fluxes) values were contained within a narrow range (Fig. 3). Interestingly, the fluxes measured above plant and water and above water-only were low and similar for the plant species with floating leaves and for *Eleocharis palustris*. Fluxes in $\text{mg CH}_4 \text{ m}^{-2} \text{ h}^{-1}$ above the plant + water and above the water-only were between 0.4 and 1 for *Nuphar lutea*; between 0 and 0.4 for *Potamogeton natans*; between 0.6 and 1.9 for *Nymphaea alba*; between 0.6 and 1.1 for *Eleocharis Palustris* (Fig. 3).

Correlations among the fraction of oxidized CH_4 , CH_4 concentration, and CH_4 fluxes

Plant-only fluxes decreased with increasing fractions of oxidized CH_4 (Fig. 4a). This relationship was still significant when observations with *Nuphar lutea*, for which plant-only fluxes could be underestimated, were removed ($\rho = -0.6$, p value = 0.001). The CH_4 plant-only flux decreased, and the fraction of oxidized CH_4 increased overall with increasing CH_4 sediment concentration (Fig. 4b–c); however, neither of these relationships was significant.

Discussion

The results show that CH_4 rhizospheric oxidation was generally higher for plants that efficiently transport gases, where up to 81% of CH_4 was oxidized (Figs. 2, S4), as hypothesized. Several studies have previously indicated that CH_4 rhizospheric oxidation can be very high but variable between stands of different plant species, affecting up to 100% of the produced CH_4 (Tyler et al. 1997; Popp et al. 1999; Zhang et al. 2016). Controlled experiments indicated that variation in CH_4 oxidation is linked to different O_2 supply rates to the rhizosphere but no generalization to other plant species than the ones studied was possible

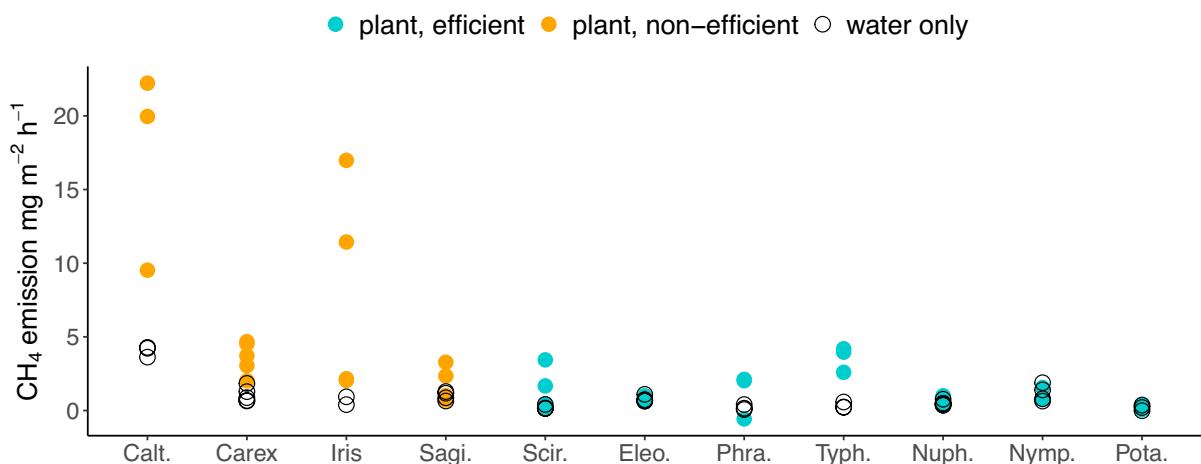
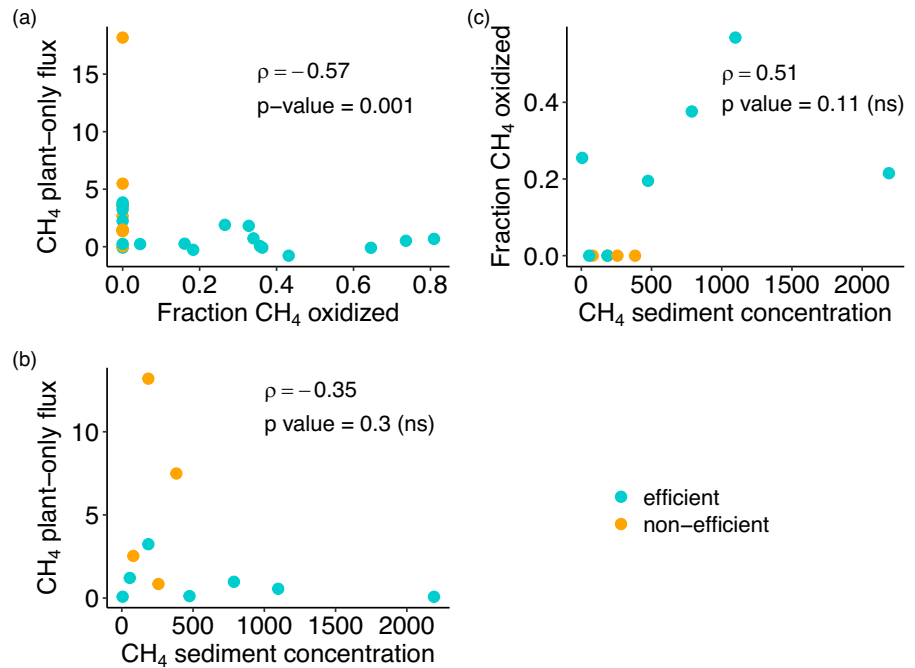


Fig. 3 Water-only (black circles) and plant and water (blue or orange points) CH_4 fluxes. Water-only fluxes refer to the diffusive fluxes of CH_4 from the water surface and are measured above the water adjacent to plants. Plant and water fluxes

include the diffusive flux of CH_4 from the water surface and the flux of CH_4 transported by plant parts present in the chambers. For plant name abbreviations see Table 1

Fig. 4 Correlations between CH₄ concentration in the sediment (in $\mu\text{mol L}^{-1}$), the fraction of oxidized CH₄ in the rhizosphere and the CH₄ transported by plants (“plant-only flux” in $\text{mg CH}_4 \text{ m}^{-2} \text{ h}^{-1}$). In **a** values are given for different individuals or groups of individuals per plant species ($n = 27$), while in **b** and **c**, values are averaged per plant species ($n = 11$). The Spearman’s rank correlation coefficients (ρ) and p-values from the Spearman tests adjusted for ties are shown in the respective panels. For plant name abbreviations, see Table 1



as the number of species and functional types investigated were low (Calhoun and King 1997; Ström et al. 2005). Accordingly, my results show that CH₄ rhizospheric oxidation is higher for several plants with high internal gas transport rates when measured in situ, potentially because they supply higher rates of O₂ to the rhizosphere (Armstrong et al. 1996; Calhoun and King 1997; Vroom et al. 2022). The aerenchyma of several leaves or stems of efficient plants contained CH₄ enriched in ¹³C and ²H in comparison to sediment pore water (Fig. S2), indicating the biological oxidation of CH₄. Several processes and mechanisms other than aerobic CH₄ oxidation can increase $\delta^{13}\text{C}$, such as the plant transport of CH₄ by diffusion (Chanton 2005), anaerobic CH₄ oxidation (Segarra et al. 2015), and CH₄ oxic production (Hartmann et al. 2020). However, by employing dual isotopes I could confirm that most of these shifts can be attributed to CH₄ oxidation, with differences in the calculations of the fraction of CH₄ being oxidized according to the $\delta^2\text{H}$ and $\delta^{13}\text{C}$ values that were mostly below 0.23 (Fig. 2).

Large variation was observed between the fraction of CH₄ oxidized calculated using $\delta^2\text{H}$ and $\delta^{13}\text{C}$ for a few samples, with sometimes very low values when using $\delta^{13}\text{C}$ (*Sagittaria sagittifolia*, *Potamogeton natans*; Fig. 2). The discrepancies between

calculations with $\delta^2\text{H}$ and $\delta^{13}\text{C}$, as well as the negative values are potentially due to the sediment signal used (Text S2) (Popp et al. 1999) or the errors related to assumptions and values used during calculations such as the fractionation factor and model used (Bastviken et al. 2002; Tsunogai et al. 2020). It is also possible that other biological processes or mechanisms occur concomitantly; for example, daytime transitions between efficient and nonefficient gas transport mechanisms due to low luminosity (Knapp and Yavitt 1992; Chanton et al. 1993, 2002; Yavitt and Knapp 1998) or reverse methanogenesis (Douglas et al. 2021) could lead to the low $\delta^{13}\text{C}$ –CH₄ signals observed in a few samples (*Potamogeton natans*, *Phragmites australis* Fig. S2). Additionally, the fraction of CH₄ oxidized sometimes greatly differs among individuals of the same species, potentially because of sediment heterogeneity close to the roots (Colmer 2003; Maisch et al. 2019), different phenological stages (Van Der Nat and Middelburg 1998b), or varying weather conditions (Knapp and Yavitt 1992; Chanton et al. 1993; Yavitt and Knapp 1998). These factors probably explain the differences when comparing datasets obtained from 2020 and 2023 (Fig. 2). Nevertheless, both the magnitude and the frequent occurrence of CH₄ oxidation I found in the rhizosphere of several efficient plants indicate that this

process is likely substantial and has an ecosystem significance for CH₄ fluxes.

Indeed, plants with a high fraction of CH₄ oxidized consistently had low plant-only CH₄ emissions (Figs. 2, 4a). When the convective bulk flow is at a steady state, outside the transition phases, CH₄ plant transport is assumed to be limited by the diffusion of CH₄ between the sediment pore water and the roots (Chanton and Dacey 1991; Kelker and Chanton 1997; Henneberg et al. 2012). Accordingly, my data indicate that CH₄ plant transport decreases because a large portion of the produced CH₄ is oxidized in the rhizosphere, likely decreasing the sediment pore-water concentration very near the roots (Fig. 4a). Variable plant-only fluxes from efficient plants or plants growing in deeper water have been reported via in situ measurements; in mg CH₄ per hour per individual: 0.1–0.6 in Sebacher et al. (1985); ca. 0 in Hamilton et al. (2014); 0–0.5 for *Typha latifolia* in Desrosiers et al. (2022) assuming a high density of 100 shoots per m²; 0.2–2.3 for water lily in Villa et al. (2020) assuming a leaf surface area of 0.05 m². However, no link has been made with CH₄ oxidation in the rhizosphere by studies reporting in situ flux measurements. My results reveal that CH₄ rhizospheric oxidation can strongly decrease net CH₄ release for some plants and may explain part of the interspecific variability that is observed in situ, as suggested by previous experimental studies (Ström et al. 2005; Kao-Kniffin et al. 2010).

CH₄ rhizospheric oxidation however did not systematically decrease net CH₄ release for all efficient plants (Fig. 4a). Ultimately, CH₄ concentration close to plant roots depends on the availability of labile organic matter for decomposition and methanogenesis, and the supply of O₂ or alternative electron acceptors (Agethen et al. 2018). In a system with a low availability of substrates for decomposition, CH₄ production may be suppressed by plant rhizospheric oxidation. Conversely, in a system with a high availability of substrates for decomposition, CH₄ oxidation may not significantly decrease overall plant CH₄ release (Agethen et al. 2018). The first was unlikely in our case because we detected high concentrations of CH₄ in the aerenchyma of all plants. However, the latter may explain the low CH₄ oxidation detected in the rhizosphere of *Typha latifolia*, which is highly productive (Wetzel 2001). Additional studies are therefore needed to confirm the causality between

CH₄ oxidation and low plant CH₄ release for different species to assess how this relationship varies in different systems and temporally, both at short and long (season) time scales.

Interestingly, low plant-only CH₄ fluxes were observed for efficient plants despite growing in a sediment with high CH₄ concentrations (Fig. 4c). In particular, for floating-leaved plants which often had a high fraction of CH₄ oxidized, CH₄ sediment pore water concentrations were consistently high (Fig. 4c). For these plants, the isotope signature of CH₄ in sediment indicated that the contribution of CO₂ reduction to CH₄ production was greater than that of CH₄ produced via acetate fermentation (Popp et al. 1999; Whiticar 1999; Figs. 1, S2) suggesting that organic matter was generally less fresh or labile than for emergent plants (Hornibrook et al. 2000). These findings indicate that floating plants were growing in sediments with different properties and suggest that floating-leaved plants have the capacity to strongly reduce CH₄ concentration in sediments, but only locally near the plant roots at “microsites” (Colmer 2003; Maisch et al. 2019; Münchberger et al. 2019). These findings imply that mechanistic models based on CH₄ concentrations or production rates from sediment sampled outside the bulk rhizosphere may poorly predict plant-only CH₄ fluxes for efficient plants, and in particular for floating-leaved plants. However, for non-efficient plants, plant-only CH₄ fluxes have been successfully related to the CH₄ concentration in sediment and its production rate (Sigren et al. 1997; Byrd et al. 2000).

This study reveals a tight link between CH₄ oxidation and plant CH₄ release, and a generally higher extent of CH₄ oxidation for plants that efficiently transport gases. Additional mechanistic studies are needed to better understand this relationship and how much sedimentary CH₄ escapes via other CH₄ transport pathways. Such work is needed to ultimately define plant functional types and traits that lead to differences in CH₄ emissions, and improve estimates and models of freshwater CH₄ emissions.

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Author contributions Charlotte Grasset: conceptualization; data curation; formal analysis; funding acquisition; investigation; methodology; and project administration.

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Data availability All the data that support the findings of this study are publicly available at Grasset (2026). <https://doi.org/10.5281/zenodo.22209912>.

Declarations

Competing interests The author declares that there are no competing interests.

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